

Heaven's Light is Our Guide

Rajshahi University of Engineering & Technology



Department of Computer Science & Engineering
23 Series

Course Code: 2206

Course Title: Microprocessor, Microcontroller & Assembly Language

Lab No.: 01

Submitted By:

Name: Md. Farhan Masud
Roll: 2303011
Section: A
Session: 2023-24

Submitted To:

Sadia Zaman Mishu
Assistant Professor
Dept. of Computer Science &
Engineering, RUET

Experiment No.: 01

Experiment Name: Taking two number inputs and showing output

Code:

```
.MODEL SMALL
.STACK 100H
.DATA
.CODE
MAIN PROC
    MOV AH,1
    INT 21H
    MOV BL,AL

    MOV AH,2
    MOV DL,'+'
    INT 21H
    MOV AH,1
    INT 21H
    ADD BL,AL

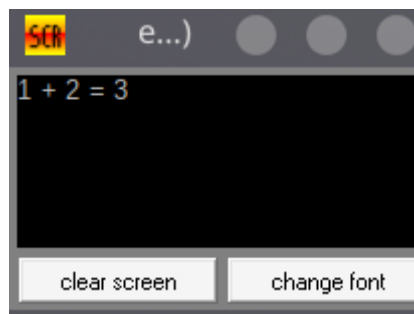
    SUB BL,'0'
    MOV AH,2

    MOV DL,'='
    INT 21H

    MOV DL,BL
    INT 21H

    MOV AH, 4CH
    INT 21H
MAIN ENDP
END MAIN
```

Output:



Discussion:

The Program successfully inputs two numbers and outputs their sum. However, the input is actually taken as character. So, we must modify the final output by subtracting the value of '0'. The program however only works up until the sum is less than 10.

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Department of Computer Science & Engineering
23 Series

Course Code: 2206

Course Title: Microprocessor, Microcontroller & Assembly Language

Lab No.: 02

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Section: A
Session: 2023-24

Submitted To:

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Experiment No.: 01

Experiment Name: Convert a lower-case letter to an upper-case letter

Code:

```
.MODEL SMALL
.STACK 100H
.DATA
CR EQU 0DH
LF EQU 0AH
MSG1 DB 'ENTER A LOWER CASE LETTER: $'
MSG2 DB CR, LF, 'IN UPPER CASE IT IS: '
CHAR DB ?, '$'
.CODE
MAIN PROC

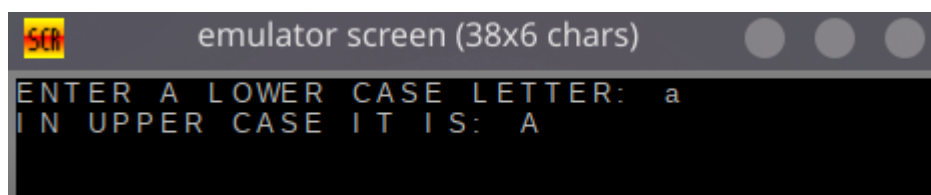
    MOV AX, @DATA
    MOV DS, AX
    LEA DX, MSG1
    MOV AH, 9
    INT 21H

    MOV AH, 1
    INT 21H
    SUB AL, 20H
    MOV CHAR, AL

    LEA DX, MSG2
    MOV AH, 9
    INT 21H

    MOV AH, 4CH
    INT 21H
MAIN ENDP
END MAIN
```

Output:



Discussion:

A simple input is given in small letter, and the output is properly shown in a new line which is the desired capital letter.

Experiment No.: 02

Experiment Name: Showing the decimal value of Hex numbers(A to F)

Code:

```
.MODEL SMALL
.STACK 100H
.DATA
.CODE
MAIN PROC

    MOV AX, @DATA
    MOV DS, AX

    MOV AH, 1
    INT 21H
    MOV BL, AL

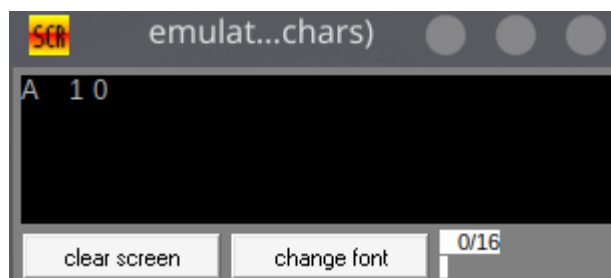
    MOV AH, 2
    MOV DL, ' '
    INT 21H

    MOV DL, '1'
    INT 21H

    SUB BL, 'A'
    ADD BL, '0'
    MOV DL, BL
    INT 21H

    MOV AH, 4CH
    INT 21H
MAIN ENDP
END MAIN
```

Output:



Discussion:

The program takes Hex inputs from 'A' to 'F' and shows the corresponding decimal output. The '1' is prewritten and we compare with 'A' to determine the 2nd digit.

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Department of Computer Science & Engineering
23 Series

Course Code: 2206

Course Title: Microprocessor, Microcontroller & Assembly Language

Lab No.: 03

Submitted By:

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Section: A
Session: 2023-24

Submitted To:

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Experiment No.: 01

Experiment Name: Input three letters while checking for correctness and whether they are capital letters and output them in an ascending order in lower-case

Code:

```
.MODEL SMALL
.STACK 100H
.DATA
A DB ?
B DB ?
C DB ?
E DB 'ERROR!$'

.CODE
MAIN PROC
    MOV AX,@DATA
    MOV DS,AX

    MOV AH,1

    INT 21H
    MOV CL,AL

    INT 21H
    MOV BL,AL

    INT 21H

    CMP AL,'A'
    JL ERROR
    CMP AL,'Z'
    JG ERROR

    CMP BL,'A'
    JL ERROR
    CMP BL,'Z'
    JG ERROR

    CMP CL,'A'
    JL ERROR
    CMP CL,'Z'
    JG ERROR

COMPARE_AB:
    CMP AL,BL
    JL COMPARE_BC
    XCHG AL,BL;BL IS NOW BIGGER

COMPARE_BC:
    CMP BL,CL
    JL COMPARE_AB_2
    XCHG CL,BL;CL IS NOW BIGGER

COMPARE_AB_2:
    CMP AL,BL
    JL PRINT
    XCHG AL,BL;BL IS NOW BIGGER

PRINT:
    MOV A,AL
    MOV AH,2
    ADD A,32
```

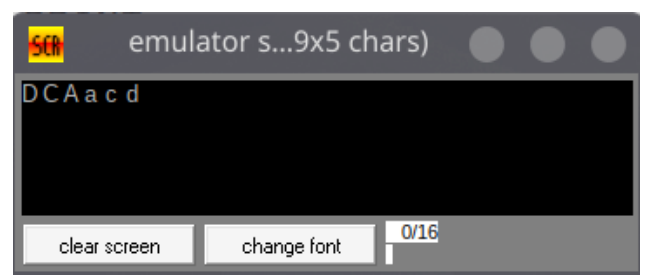
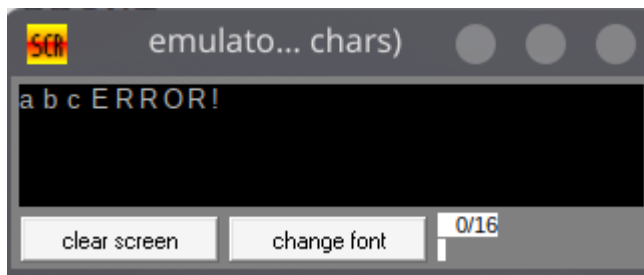
```

MOV DL,A
INT 21H
ADD BL,32
MOV DL,BL
INT 21H
ADD CL,32
MOV DL,CL
INT 21H
JMP ENDING
ERROR:
MOV AH,9
LEA DX,E
INT 21H

ENDING:
        MAIN ENDP
END MAIN

```

Output:



Discussion:

The program successfully checks the inputs for errors and prints the output in an ascending manner as well as in lower-case.

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Department of Computer Science & Engineering
23 Series

Course Code: 2206

Course Title: Microprocessor, Microcontroller & Assembly Language

Lab No.: 04

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Experiment No.: 01

Experiment Name: Odd-Even checking with switch-case and error checking

Code:

```
.MODEL SMALL
.STACK 100H
.DATA
E DB 10,13,'EVEN',10,13,'$'
O DB 10,13,'ODD',10,13,'$'
D DB 10,13,'DECIMAL:', '$'
ERR DB 10,13,'WRONG INPUT SELECT Y/N',10,13,'$'
ASK DB 10,13,'INPUT_HEX: ', '$'
ASK_AGAIN DB 10,13,'RECHECK: ', '$'
.CODE
MAIN PROC
    MOV AX,@DATA
    MOV DS,AX

INPUT:
    MOV AH,9
    LEA DX, ASK
    INT 21H

    MOV AH,1
    INT 21H

    MOV BL, AL
    MOV AH,9
    LEA DX,D
    INT 21H
    MOV AL,BL

    CMP BL,'A'
    JL PRINT
    SUB BL,'A'
    ADD BL,'0'

    MOV AH,2
    MOV DL,'1'
    INT 21H

PRINT:
    MOV AH,2
    MOV DL,BL
    INT 21H

CHECK_1:
    AND BL,1
    CMP BL,1
    JNE EVEN
    MOV AH,9
    LEA DX,O
    INT 21H
    JMP CHECK_2
EVEN:
    MOV AH,9
    LEA DX,E
    INT 21H
CHECK_2:
    MOV AH,9
    LEA DX, ASK_AGAIN
    INT 21H
    MOV AH,1
```

```

INT 21H

CMP AL, 'Y'
JE INPUT
CMP AL, 'y'
JE INPUT

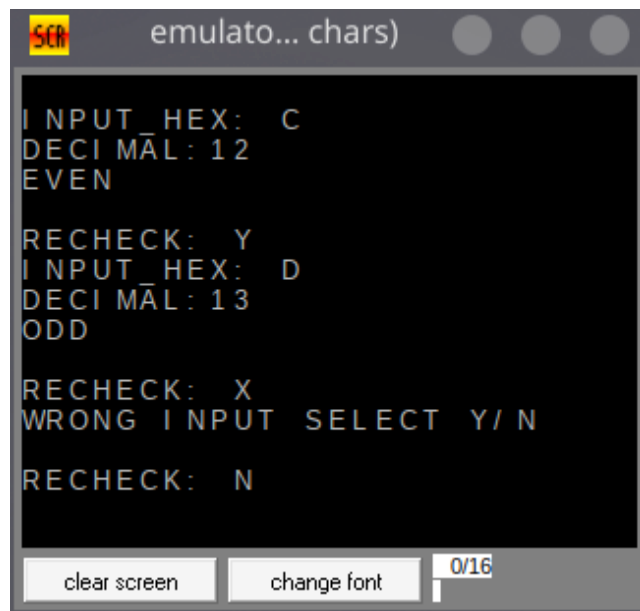
CMP AL, 'N'
JE END
CMP AL, 'n'
JE END

MOV AH, 9
LEA DX, ERR
INT 21H
JMP CHECK_2

END:
MOV AH, 4CH
INT 21H
MAIN ENDP
END MAIN

```

Output:



Discussion:

The program takes a Hex input from 0 to 'A', properly outputs the decimal value and finally shows whether it's odd or even based on AND operation.

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Department of Computer Science & Engineering
23 Series

Course Code: 2206

Course Title: Microprocessor, Microcontroller & Assembly Language

Lab No.: 05

Submitted By:

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Experiment No.: 01

Experiment Name: Binary Input-Output, Reversal, Compliment and Odd Parity Check

Code:

```
.MODEL SMALL
.STACK 100H
.DATA
ORIG DB ?
COMP DB ?
REV DB ?
ODD DB ?
TEMP DB ?

.CODE

MAIN PROC
    MOV AX,@DATA
    MOV DS,AX

    MOV BL,0
    MOV AH,1
    MOV CX,8
INPUT:
    INT 21H
    SUB AL,'0'
    SHL BL,1
    OR BL,AL
    LOOP INPUT
    JMP COMPLEMENT
COMPLEMENT:
    MOV DL,BL
    MOV ORIG,BL
    XOR DL,11111111B
    MOV COMP,DL

    MOV CX,8
    MOV AL,0
REVERSE:
    MOV TEMP,1
    AND TEMP,DL
    ROR DL,1
    SHL AL,1
    OR AL,TEMP
    LOOP REVERSE
    MOV REV,AL
    MOV CX,4
    MOV DH,0
PARITY:
    MOV TEMP,1
    AND TEMP,AL
    SHR AL,1
    XOR DH,TEMP
    LOOP PARITY
    XOR DH,1
    MOV ODD,DH

END:
    MOV AH,2

    MOV CX,8
    MOV DL,10
    INT 21H
    MOV DL,13
```

```

    INT 21H
    MOV BL,ORIG
LOOP1:
    MOV TEMP,10000000B
    AND TEMP,BL
    ROL BL,1
    ROL TEMP,1
    ADD TEMP,'0'
    MOV DL,TEMP
    INT 21H
    LOOP LOOP1

    MOV CX,8
    MOV DL,10
    INT 21H
    MOV DL,13
    INT 21H
    MOV BL,COMP
LOOP2:
    MOV TEMP,10000000B
    AND TEMP,BL
    ROL BL,1
    ROL TEMP,1
    ADD TEMP,'0'
    MOV DL,TEMP
    INT 21H
    LOOP LOOP2

    MOV CX,8
    MOV DL,10
    INT 21H
    MOV DL,13
    INT 21H
    MOV BL,REV
LOOP3:
    MOV TEMP,10000000B
    AND TEMP,BL
    ROL BL,1
    ROL TEMP,1
    ADD TEMP,'0'
    MOV DL,TEMP
    INT 21H
    LOOP LOOP3

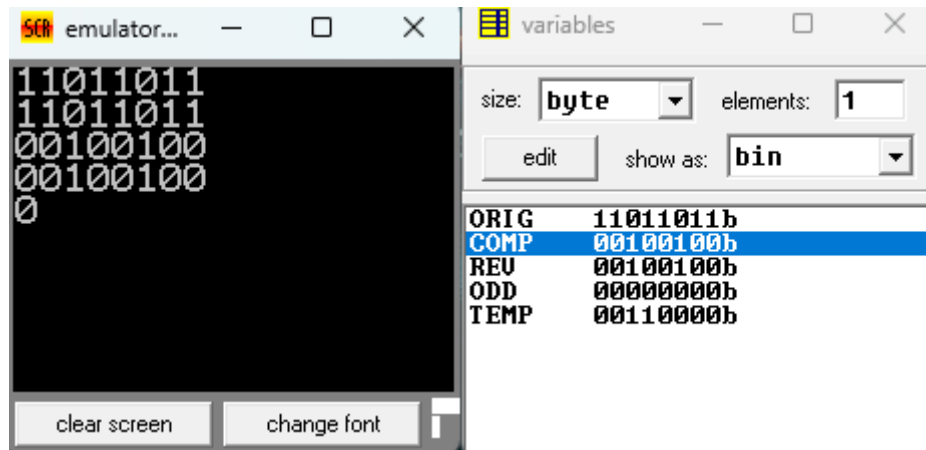
    MOV DL,10
    INT 21H
    MOV DL,13
    INT 21H
    MOV BL,REV

    MOV DL,ODD
    ADD DL,'0'
    INT 21H

    MOV AH,4CH
    INT 21H
    MAIN ENDP
END MAIN

```

Output:



Discussion:

The program-

1. Took ascii input of 0s and 1s and transformed into real binary code stored in the *ORIG* variable
2. Performed 1's compliment with bitwise arithmetic, which is showed in *COMP* variable
3. Reversed the complimented value with SHL and RHL along with bitwise arithmetic and it is in *REV* variable.
4. Lastly, we checked for odd parity which is 0 for this input in *ODD*.

Limitations-

1. Some operations could have been simplified by using carry and RCL/RCR operations
2. The use of auxiliary variables to store the result of bit-checking with AND could have been avoided with TEST opcode

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Department of Computer Science & Engineering
23 Series

Course Code: 2206

Course Title: Microprocessor, Microcontroller & Assembly Language

Lab No.: 06

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Submitted To:

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Experiment No.: 01

Experiment Name: Word based reversal

Code:

```
.MODEL SMALL
.STACK 100H
.DATA
    ARR DW 100DUP(0)

.CODE
MAIN PROC
    MOV AX,@DATA
    MOV DS,AX
    MOV SI,0
    MOV CX,0
    MOV BX,0
INPUT:
    MOV AH,1
    INT 21H
    CMP AL,0DH
    JE NEXT
    CALL COUNT
    INC CX
    MOV AH,0
    PUSH AX
    JMP INPUT
NEXT:
    POP AX
    MOV ARR[SI],AX
    ADD SI,2
    LOOP NEXT
    MOV AH,2
    MOV DX,0DH
    INT 21H
    MOV DX,0AH
    INT 21H
    MOV CX,0
PRINT:
    SUB SI,2
    MOV DX,ARR[SI]
    CMP ARR[SI],' '
    JE PRINT_STACK
    JE PRINT_
    PUSH DX
    INC CX
    CMP SI,0
    JG PRINT
    JL END
PRINT_STACK:
    POP DX
    MOV AH,2
    INT 21H
    LOOP PRINT_STACK
    MOV DX,' '
    INT 21H
    JMP PRINT

END:
    MOV AH,2
    MOV DX,0DH
    INT 21H
    MOV DX,0AH
```

```

INT 21H
MOV DX,BX
ADD DX,'0'
INT 21H
MOV AH,4CH
INT 21H
MAIN ENDP

```

```

COUNT PROC
    CMP AL,'a'
    JE YES
    CMP AL,'e'
    JE YES
    CMP AL,'i'
    JE YES
    CMP AL,'o'
    JE YES
    CMP AL,'u'
    JE YES
NO:
    RET
YES:
    INC BX
    RET
COUNT ENDP
END MAIN

```

Output:



```

this is a test
siht si a tset
4

```

Discussion:

The program uses the main stack and an array as a secondary stack. Firstly, the entire input is put into stack and then from the stack to the array. After that word by word is reversed with the help of the two stacks. The program successfully reversed the string word by word.

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Rajshahi University of Engineering & Technology



Department of Computer Science & Engineering
23 Series

Course Code: 2206

Course Title: Microprocessor, Microcontroller & Assembly Language

Lab No.: 07

Submitted By:

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Section: A
Session: 2023-24

Submitted To:

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Experiment No.: 01

Experiment Name: Time division

Code:

```
.MODEL SMALL
.STACK 100H
.DATA
INPUT DW ?
INFO DB 'GIVE TIME INPUT: $'
HR DB 'HR: $'
MIN DB 'MIN: $'
SEC DB 'SEC: $'
.CODE
MAIN PROC
    MOV AX,@DATA
    MOV DS,AX
    MOV AH,9
    LEA DX,INFO
    INT 21H

    CALL INDEC
    MOV INPUT,AX
    CALL NL
    MOV CX,3

MAIN_LOOP:
    MOV BX,60
    MOV DX,0 ; HR
    DIV BX
    PUSH DX
    LOOP MAIN_LOOP

PRINT:
    MOV AH,9
    LEA DX, HR
    INT 21H
    POP BX
    CALL OUTDEC
    CALL NL

    MOV AH,9
    LEA DX, MIN
    INT 21H
    POP BX
    CALL OUTDEC
    CALL NL

    MOV AH,9
    LEA DX, SEC
    INT 21H
    POP BX
    CALL OUTDEC
    MOV AH,4CH
    INT 21H
    MAIN ENDP

INDEC PROC
    PUSH BX
    PUSH CX
    PUSH DX
    MOV BX,0
    MOV CX,10
INPUT_LOOP:
```

```

        MOV AH,1
        INT 21H
        CMP AL,' '
        JE ENDIT
        CMP AL,0DH
        JE ENDIT
        AND AX,0FH
        XCHG AX,BX
        MUL CX
        ADD AX,BX
        MOV BX,AX
        JMP INPUT_LOOP
ENDIT:
        MOV AX,BX
        POP DX
        POP CX
        POP BX
        RET
INDEC ENDP

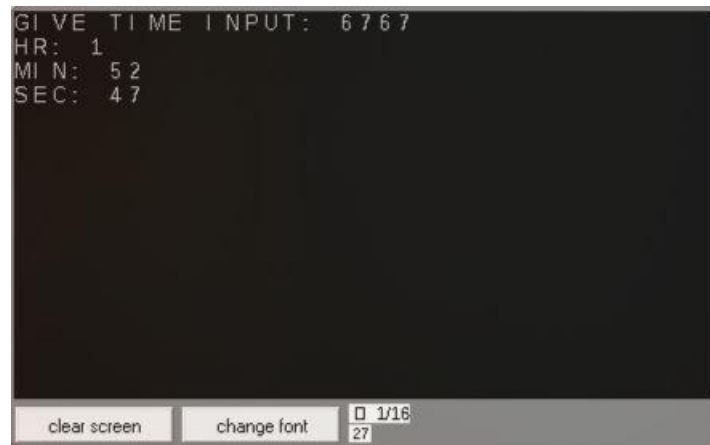
OUTDEC PROC
        PUSH AX
        PUSH DX
        PUSH CX
        MOV CX,0
        MOV AX,BX
        MOV BX,10
STACK_LOAD:
        MOV DX,0
        DIV BX
        PUSH DX
        INC CX
        CMP AX,0
        JE STACK_UNLOAD
        JMP STACK_LOAD
STACK_UNLOAD:
        MOV AH,2
        POP DX
        ADD DL,'0'
        INT 21H
        LOOP STACK_UNLOAD
        POP CX
        POP DX
        POP AX
        RET
OUTDEC ENDP

NL PROC
        PUSH DX
        PUSH AX
        MOV AH,2
        MOV DL,0AH
        INT 21H
        MOV DL,0DH
        INT 21H
        POP AX
        POP DX
        RET
NL ENDP
END MAIN

```

Output:

```
GIVE TIME INPUT: 6767
HR: 1
MI N: 52
SEC: 47
```



Discussion:

The program implemented decimal input and output with custom procedures using stack, DIV & MUL operators. After taking input in decimal it divides the value by 60 three times to determine hour, minute and second values. Using the output proc it emits the values after each division.

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Department of Computer Science & Engineering
23 Series

Course Code: 2206

Course Title: Microprocessor, Microcontroller & Assembly Language

Lab No.: 08

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Section: A
Session: 2023-24

Submitted To:

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Experiment No.: 01

Experiment Name: Palindrome Checker

Code:

```
.MODEL SMALL
.STACK 100H
.DATA
ARR DB 50 DUP('$')
PASSED DB 'PALINDROME$'
FAILED DB 'NOT PALINDROME$'
.CODE
MAIN PROC
    MOV AX,@DATA
    MOV DS,AX

    CALL INPUT
    CALL NL

    LEA DI,ARR

    MOV AX,DI ; POINTER SWAP
    XCHG SI,AX
    MOV DI,AX
CHECK:
    MOV AX,SI
    CMP AX,DI
    JGE PASS

    MOV AL,[SI]
    CMP AL,[DI]
    JNE FAIL
    INC SI
    DEC DI
    JMP CHECK
PASS:
    MOV AH,9
    LEA DX,PASSED
    INT 21H
    JMP ENDIT
FAIL:
    MOV AH,9
    LEA DX,FAILED
    INT 21H
    JMP ENDIT
ENDIT:
    MOV AH,4CH
    INT 21H
    MAIN ENDP

INPUT PROC
    PUSH AX
    PUSH BX
    PUSH CX
    PUSH DX
    LEA SI,ARR
    MOV CX,0
    MOV AH,1
INPUT_LOOP:
    INT 21H
    CMP AL,0DH
    JE INPUT_END
```

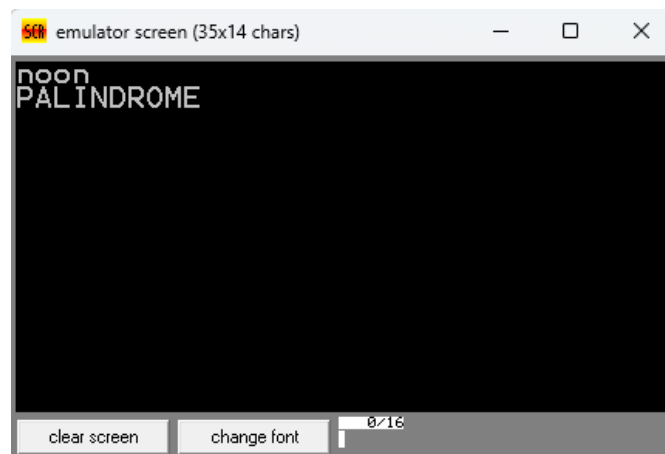
```

MOV [SI],AL
INC SI
JMP INPUT_LOOP

INPUT_END:
DEC SI; POINTS TO THE LAST ELEMENT
POP DX
POP CX
POP BX
POP AX
RET
INPUT ENDP
NL PROC
PUSH AX
PUSH DX
MOV AH,2
MOV DL,0DH
INT 21H
MOV DL,0AH
INT 21H
POP DX
POP AX
RET
NL ENDP
END MAIN

```

Output:



Discussion:

The program checked for palindrome of a string by using SI & DI pointers and traversing SI forward and DI backward till SI ≥ DI. If the content in SI & DI don't match, it's not a palindrome, otherwise it is.